

Lab 1. Oracle Database installation, connection and configuration

Content

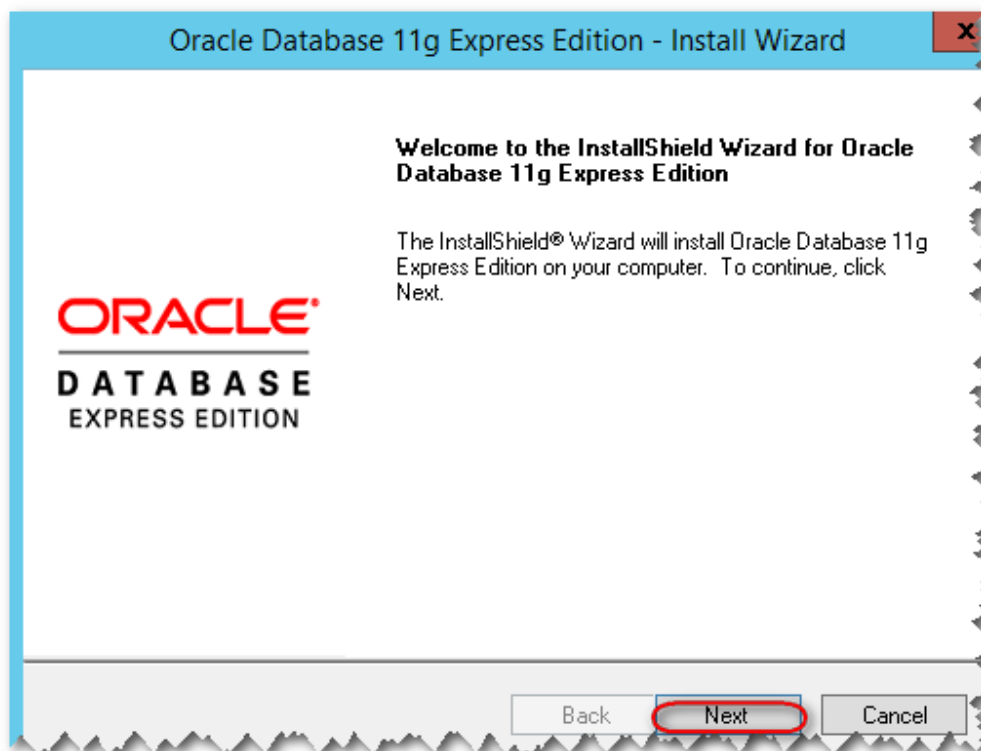
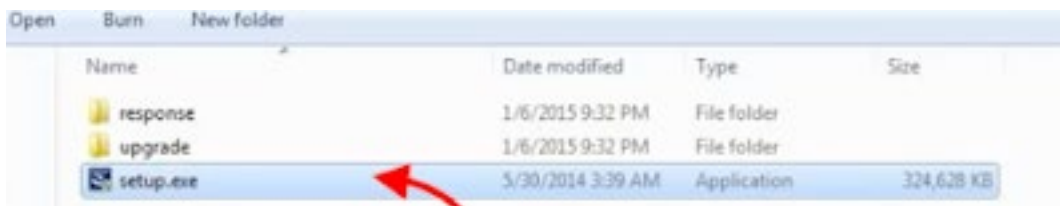
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Oracle Database Express Edition installation

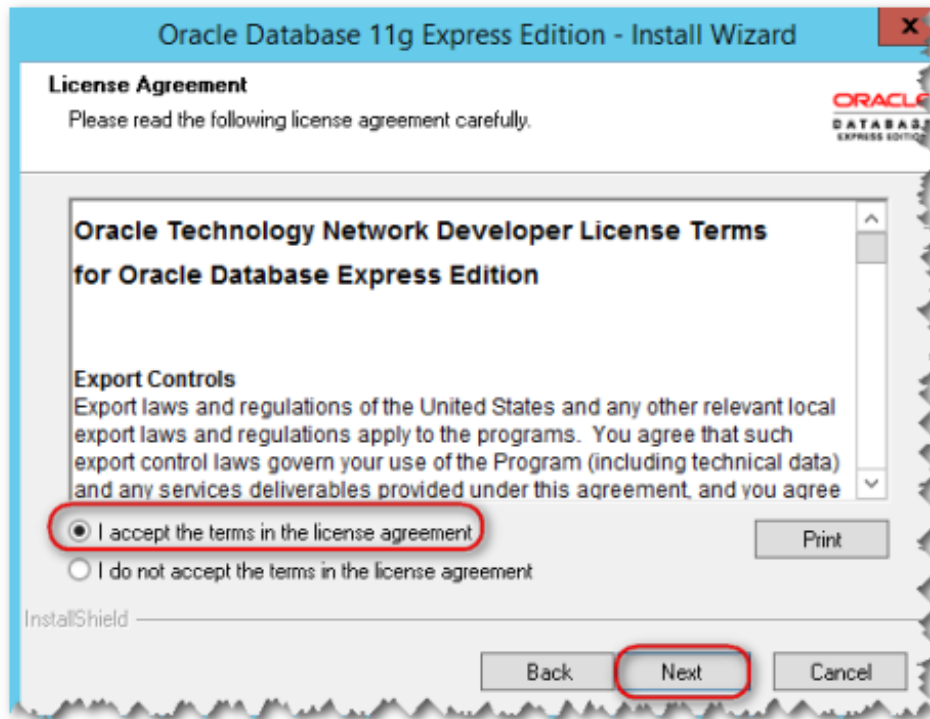
1. Download the *Oracle DB XE installer* (~320MB):

https://drive.google.com/drive/folders/1WbjAVT0UTtqwjGY9RTd_tC4tT75MEs5h

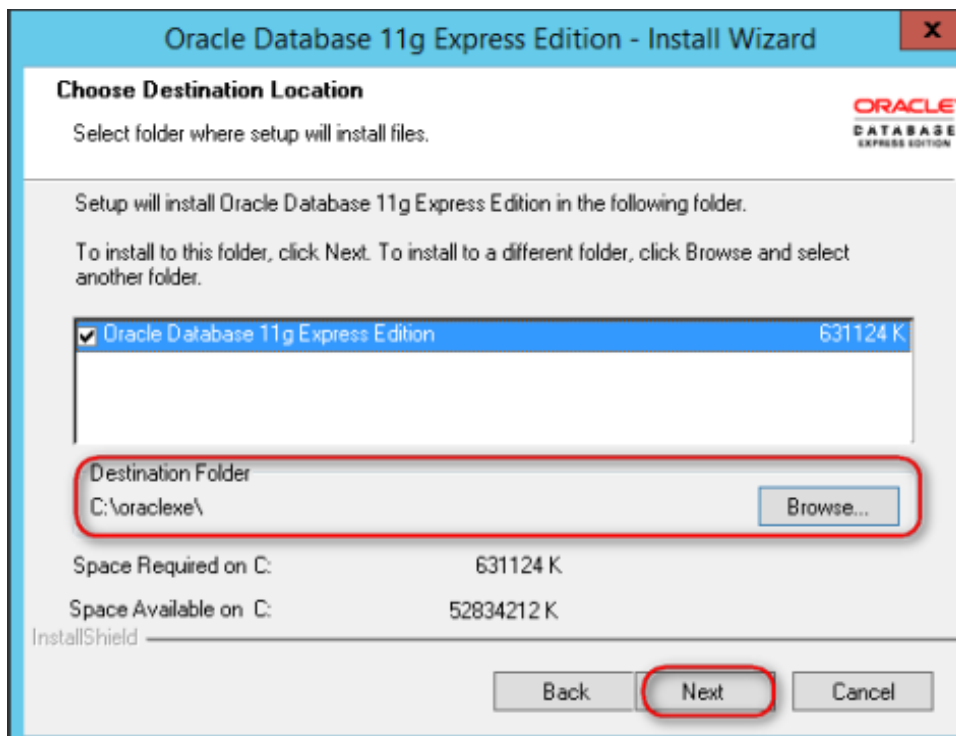
2. Run **setup.exe** and click **Next**



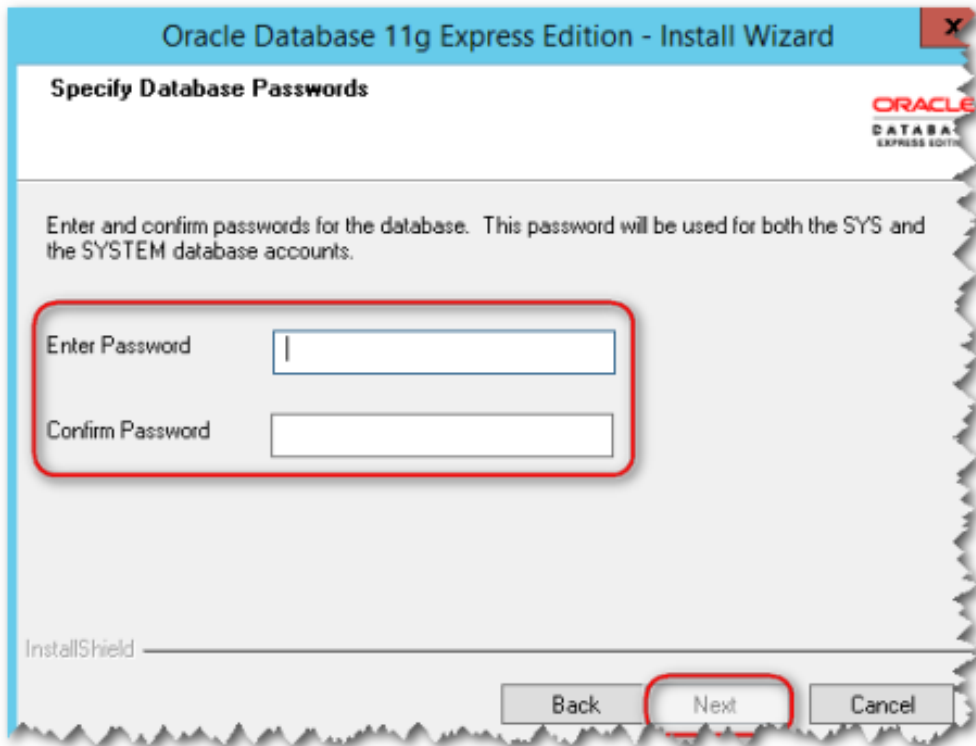
3. Accept the *license agreement* and click **Next**



4. Choose the *location* for installing Oracle DB and click **Next**



5. Configure the *password* for the database. Save this password because you need it to connect with the database.



Oracle Database 11g Express Edition - Install Wizard

Specify Database Passwords

Enter and confirm passwords for the database. This password will be used for both the SYS and the SYSTEM database accounts.

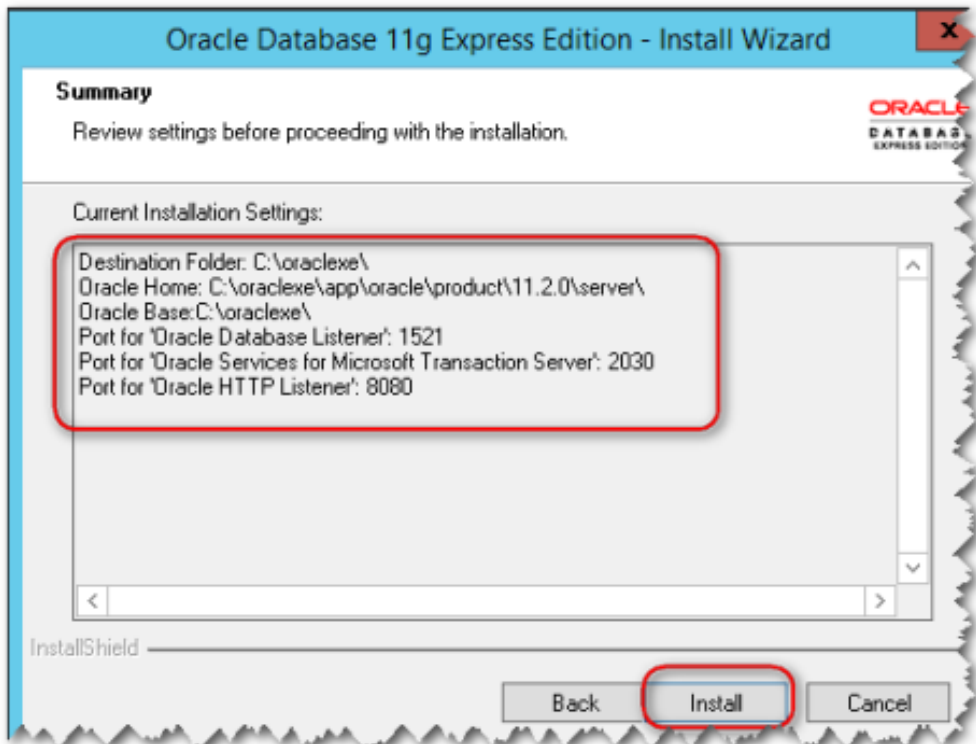
Enter Password

Confirm Password

InstallShield

Back Next Cancel

6. Save the *summary*. These are the details for connecting with the database.



Oracle Database 11g Express Edition - Install Wizard

Summary

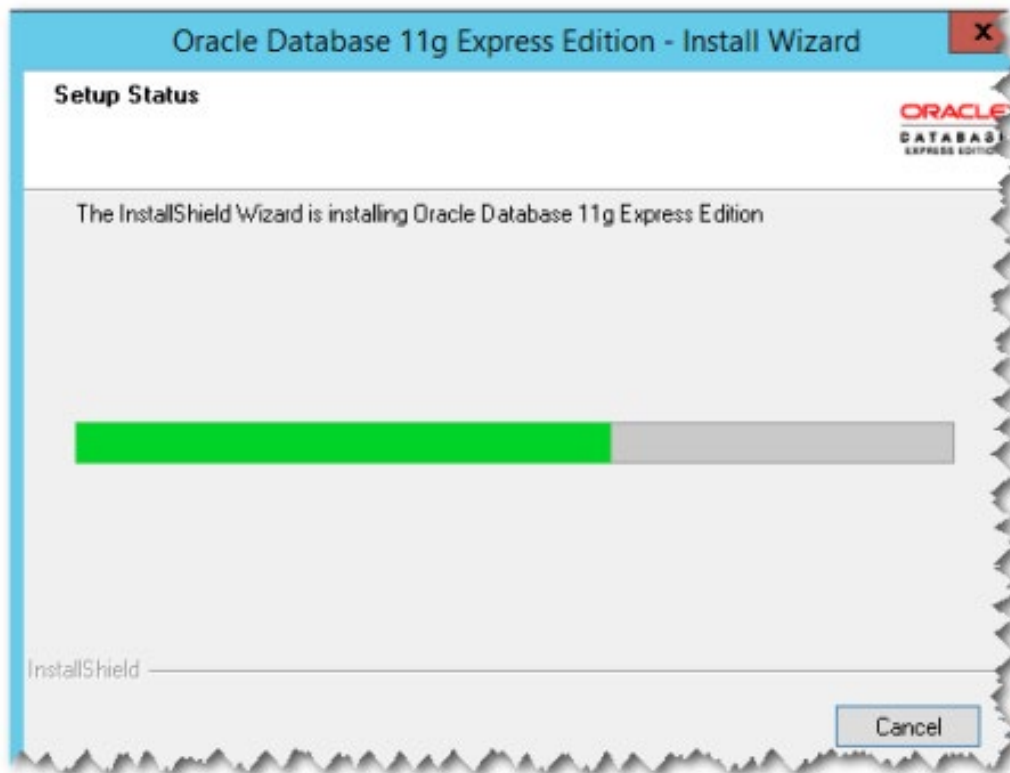
Review settings before proceeding with the installation.

Current Installation Settings:

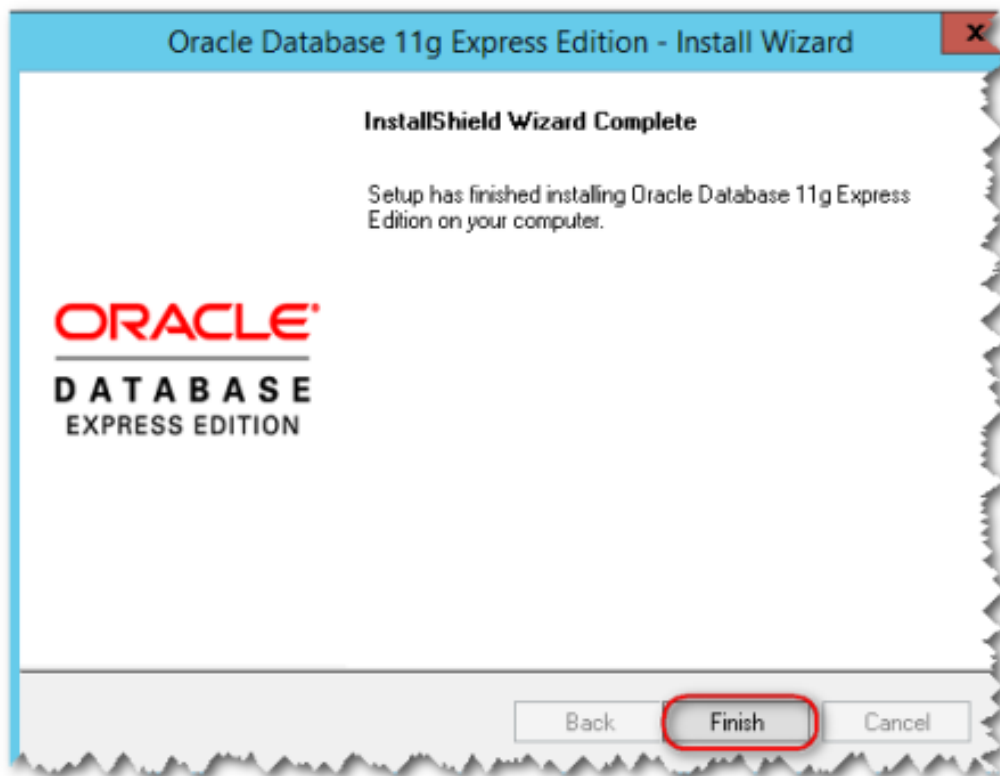
Destination Folder: C:\oraclexe\
Oracle Home: C:\oraclexe\app\oracle\product\11.2.0\server\
Oracle Base: C:\oraclexe\
Port for 'Oracle Database Listener': 1521
Port for 'Oracle Services for Microsoft Transaction Server': 2030
Port for 'Oracle HTTP Listener': 8080

InstallShield

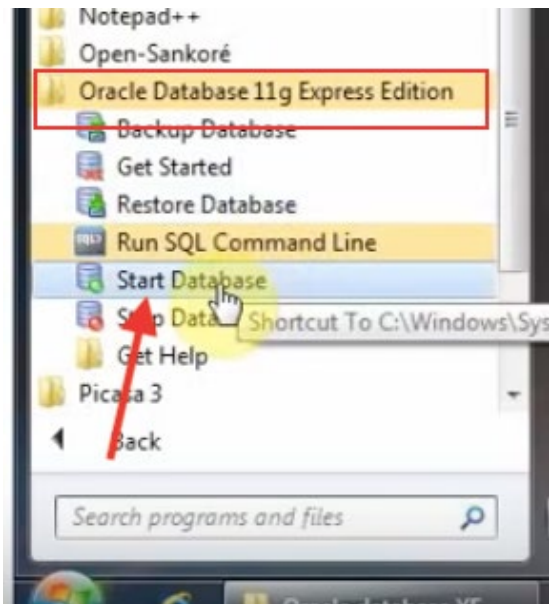
Back Install Cancel



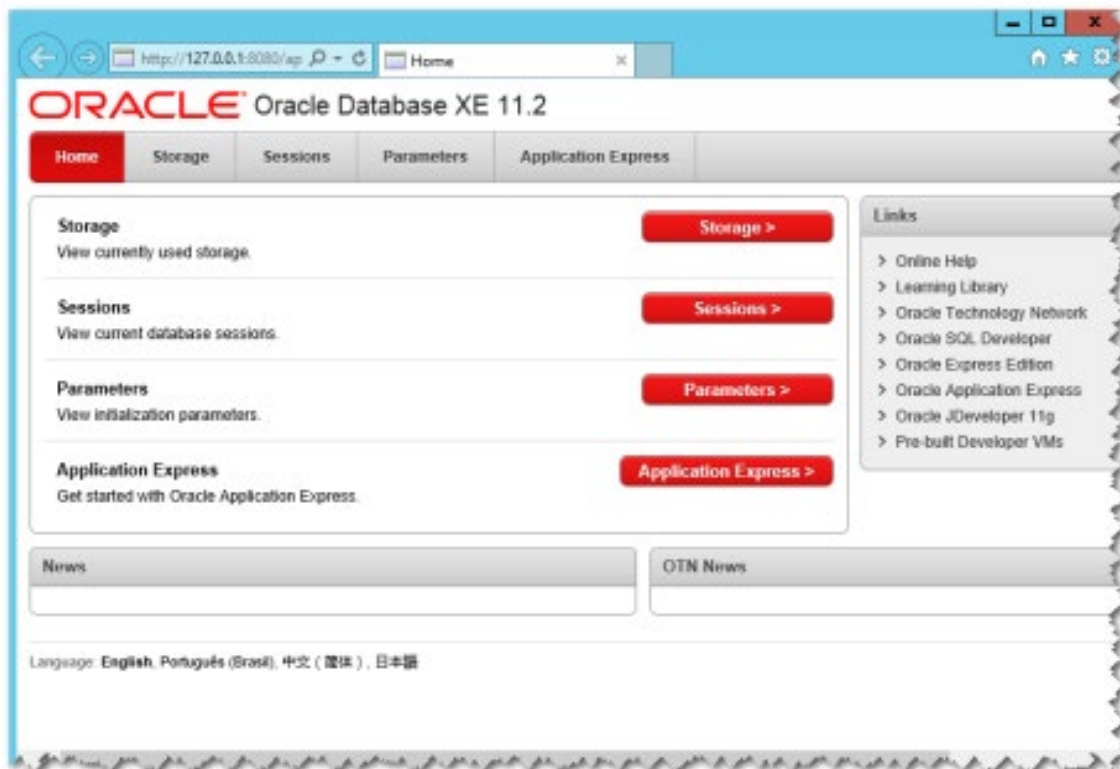
7. Database installation is **successful**. Now, the database is ready to use.



8. Mostly, the Oracle database gets started automatically after installation. But as this is the first time, let's start the database manually. Search for the Oracle Database 11g XE and click on **Start Database**.



9. Once installed, you can run Oracle on the Web at:
<http://127.0.0.1:8080/apex/f?p=4950:1:535275793956732>



Local database connection

Establishing a connection locally means running the *SQL command-line utility on the same computer* where Oracle Database 11g XE is installed, and initiating a database connection command using the valid database credentials.

The following screenshot is taken from Oracle Database 11g XE on Windows, and demonstrates establishing a local database connection to the database:

```
C:\>cd c:\oraclexe\app\oracle\product\11.2.0\server\bin
c:\oraclexe\app\oracle\product\11.2.0\server\bin>
c:\oraclexe\app\oracle\product\11.2.0\server\bin>sqlplus "/nolog"
SQL*Plus: Release 11.2.0.2.0 Production on Wed Dec 21 00:03:58 2011
Copyright (c) 1982, 2010, Oracle. All rights reserved.
SQL>
SQL> connect system/sys1234
Connected.
SQL>
```

ORA_DBA is a local Windows group that is *automatically* created when you install Oracle Database XE, and your Windows username is automatically added to this group. Members of *ORA_DBA* can connect to the Oracle Database XE without a password, as shown in the following screenshot:

```
C:\Users\amomen>sqlplus "/as sysdba"
SQL*Plus: Release 11.2.0.2.0 Production on Tue May 1 13:09:48 2012
Copyright (c) 1982, 2010, Oracle. All rights reserved.

Connected to:
Oracle Database 11g Express Edition Release 11.2.0.2.0 - Production
SQL>
```

Oracle Net Listener

To connect to the database remotely, Oracle Net Listener should be running on the host where Oracle Database 11g XE is installed. When a remote client initiates a database connection request, this connection request is received by Oracle Net Listener. The job of the listener is to listen to these incoming database connection requests and hand over the connection requests to the appropriate databases.

Thereafter, the remote-client connection directly communicates with the database without the need for the listener. Without the listener service running, we cannot connect to the database remotely. There may be more than one database running on the host server; however, you will only have one database listener for all incoming requests. Based on the incoming request, Oracle Net Listener will hand over the connection request to the appropriate database.

Oracle Net Listener and the `tnsnames.ora` files are configured by default when we install Oracle Database 11g XE. The listener's configuration file is located under `$ORACLE_HOME\network\admin\listener.ora`. Oracle Database XE automatically creates a Windows service for Oracle Net Listener (OracleXETNSListener).

Configuring Oracle Net Listener

`listener.ora` is the name of the Oracle Net Listener configuration file, and it resides under `$ORACLE_HOME\network\admin` folder.

The content of a sample `listener.ora` file in its simplest form is as follows:

```
SID_LIST_LISTENER = (SID_LIST =
    (SID_DESC =
        (SID_NAME = XE)
        (ORACLE_HOME = C:\oraclexe\app\oracle\product\11.2.0\server)
    )
)
LISTENER = (DESCRIPTION_LIST =
    (DESCRIPTION =
        (ADDRESS = (PROTOCOL = TCP)(HOST = amomen-PC)(PORT = 1521))
    )
)
```

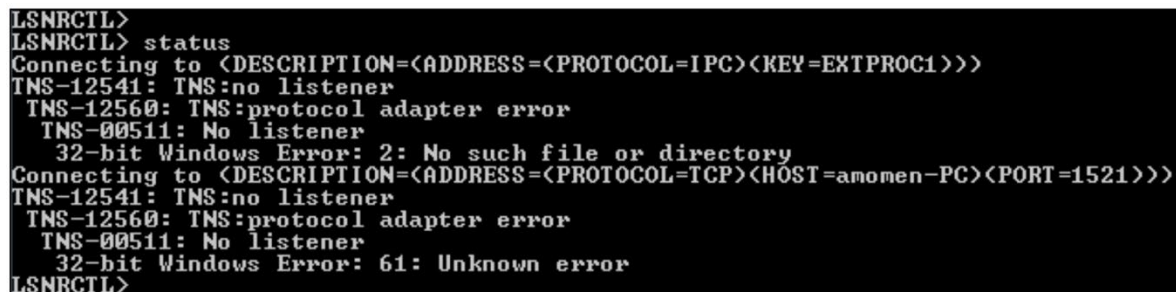
The description of the parameters is as follows:

- LISTENER: The name of the listener
- SID_NAME: The Oracle Database name
- ORACLE_HOME: The Oracle software installation home directory
- HOST: The name of the host where Oracle Net Listener is running
- PORT: The port on which Oracle Net Listener is listening to the incoming requests

Make the necessary changes to the preceding parameters, save the file, and restart the Windows Oracle Listener service OracleXETNSListener.

Viewing the status of Oracle Net Listener

If the listener is not started, the command displays an error message, as shown in the following screenshot:



```
LSNRCTL>
LSNRCTL> status
Connecting to <DESCRIPTION=(ADDRESS=(PROTOCOL=IPC)<KEY=EXTPROC1>>>
TNS-12541: TNS:no listener
TNS-12560: TNS:protocol adapter error
TNS-00511: No listener
32-bit Windows Error: 2: No such file or directory
Connecting to <DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=amomen-PC)(PORT=1521))>
TNS-12541: TNS:no listener
TNS-12560: TNS:protocol adapter error
TNS-00511: No listener
32-bit Windows Error: 61: Unknown error
LSNRCTL>
```


If the listener is running, the command displays detailed listener information as shown in the following screenshot:

```
LSNRCTL>
LSNRCTL> status
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=IPC)(KEY=EXTPROC1)))
STATUS of the LISTENER
-----
Alias                     LISTENER
Version                   TNSLSNR for 32-bit Windows: Version 11.2.0.2.0 - Production
Start Date                21-DEC-2011 01:41:30
Uptime                    0 days 0 hr. 23 min. 6 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Default Service           XE
Listener Parameter File   C:\oracle\app\oracle\product\11.2.0\server\network\admin\listener.ora
Listener Log File         C:\oracle\app\oracle\diag\tnslsnr\amomen-PC\listener\alert\log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(PIPENAME=\\.\pipe\EXTPROC1ipc)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=amomen-PC)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=amomen-PC)(PORT=8080))(Presentation=HTTP)(Session=RAW))
Services Summary...
Service "CLRExtProc" has 1 instance(s).
  Instance "CLRExtProc", status UNKNOWN, has 1 handler(s) for this service...
Service "PLSExtProc" has 1 instance(s).
  Instance "PLSExtProc", status UNKNOWN, has 1 handler(s) for this service...
Service "XEKDB" has 1 instance(s).
  Instance "xe", status READY, has 1 handler(s) for this service...
Service "xe" has 1 instance(s).
  Instance "xe", status READY, has 1 handler(s) for this service...
The command completed successfully
LSNRCTL>
```

The listener is automatically configured when we install Oracle Database 11g XE.

Starting and stopping the listener

- To stop the listener on Windows, launch the Services window by navigating to Start | Control Panel | Administrative Tools | Services and stop the OracleXETNSListener service, or launch the listener utility (lsnrctl) and enter the LSNRCTL> stop command. To stop the listener on Linux, within the listener utility (lsnrctl) enter the stop command as we did in the Windows environment.
- To start the listener on Windows, launch the Services window by navigating to Start | Control Panel | Administrative Tools | Services and start the OracleXETNSListener service, or launch the listener utility (lsnrctl) and enter the LSNRCTL> start command. To start the listener on Linux, within the listener utility (lsnrctl) enter the start command as we did in Windows environment.

Configuring the tnsnames.ora file

Transparent Network Substrate (TNS) handles all remote database connections. Oracle software reads the TNS connection string to understand how to connect to the remote databases. Every Oracle database and the Oracle Client software will have this file. By default the tnsnames.ora file is located under \$ORACLE_HOME\network\admin. A sample alias entry from the tnsnames.ora file is as follows:

```
MY_XE = (DESCRIPTION =
        (ADDRESS = (PROTOCOL = TCP)(HOST = amomen-PC)(PORT = 1521))
        (CONNECT_DATA = (SERVER = DEDICATED) (SID = XE))
)
```


In the preceding code:

- MY_XE is an alias name. You can name it anything you like.
- HOST is the hostname or IP-Address where database is running.
- PORT is the port number on which Oracle Net Listener is listening.
- SID is the Oracle database name.

We will use **my_xe** as a connect string when connecting to the remote database.

Remote database connections

Establishing a connection remotely means running the *SQL command-line utility* on a different host other than the Oracle Database 11g XE machine and initiating a database connection command using the valid database credentials over the network using a connect string.

The following screenshot is taken from Oracle Database 11g XE on Windows, and demonstrates establishing a remote database connection to the database:

```
c:\>
c:\>cd c:\oraclexe\app\oracle\product\11.2.0\server\bin

c:\oraclexe\app\oracle\product\11.2.0\server\bin>
c:\oraclexe\app\oracle\product\11.2.0\server\bin>sqlplus "/nolog"

SQL*Plus: Release 11.2.0.2.0 Production on Wed Dec 21 00:31:15 2011
Copyright (c) 1982, 2010, Oracle. All rights reserved.

SQL>
SQL> connect system/sys1234@xe
Connected.
SQL>
```

As discussed earlier, a connect string defines which database server to connect to, on what port, and using what protocol. In the preceding example, **xe** is the name of the *connect string*.

Alternatively, we can connect to the database by providing the database *hostname* and *port* as shown in the following screenshot:

```
SQL>
SQL> connect system/sys1234@amomen-pc:1521
Connected.
SQL>
```

The database hostname name and port information are encapsulated in the connect string.

Tài liệu tham khảo

- [1] <https://www.oicbasics.com/2020/01/download-oracle-database-11g-xe-express.html>
- [2] <https://dangvinhcuong.wordpress.com/2015/04/15/oraclelesson01-cai-dat-oracle-xeoracle-database-express-edition-11g/>
- [3] <https://docs.oracle.com/en/database/oracle/oracle-database/21/xeinw/installing-oracle-database-xe.html#GUID-7BF9ACDC-1281-494C-AF96-77A25B1BD67D>
- [4] <https://drive.google.com/file/d/1B3v1dUHPV0Tu82WWZJG9Va8iX5w4tWL2/view?usp=sharing>
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- [6] https://docs.oracle.com/cd/E17781_01/server.112/e18804/getstart.htm#ADMQS110
- [7] <https://www.oracle.com/database/technologies/xe-prior-release-downloads.html> (11g)
- [8] <https://www.oracle.com/database/technologies/express-edition-downloads.html> (18c)
- [9] <https://www.oracle.com/in/database/technologies/instant-client/macos-intel-x86-downloads.html>